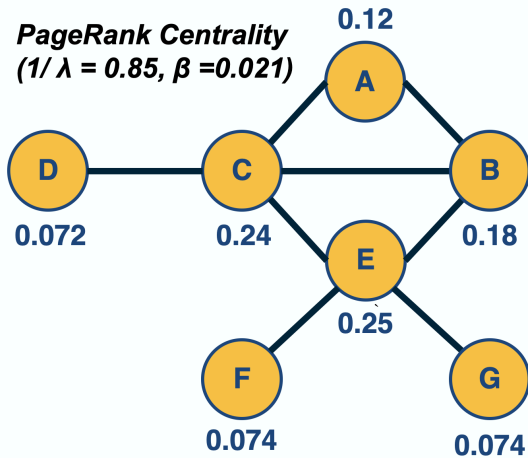


L6: PageRank Centrality

PageRank Centrality
($1/\lambda = 0.85$, $\beta = 0.021$)



PageRank, created by the co-founder of Google, Larry Page, is a famous centrality metric because it was the main algorithm that made Google the most successful search engine back in the late 1990s.

In fact, PageRank is only a slight modification of the Katz centrality metric. It is based on the following idea: if a node j points to a node i (and thus $A_{i,j} = 1$), and node j has $k_{j,out}$ outgoing connections, then the centrality of node j should be "split" among those $k_{j,out}$ neighbors. In other words, the "wealth" of node j should not be just inherited by all nodes it points to – but rather, the "wealth" of node j should be split among those nodes.

So, the defining equation of PageRank centrality becomes:

$$v_i = \beta + \frac{1}{\lambda} \sum_j A_{i,j} \frac{v_j}{k_{j,out}}$$

where the summation is over all nodes j that point to i (and thus, $k_{j,out}$ is non-zero).

Please contrast this equation with the definition of Katz centrality.

In matrix form, similar to Katz centrality, the previous definition becomes (when $\beta = 1$):

$$v = (I - \frac{1}{\lambda} A D)^{-1} \cdot \mathbf{1}$$

where D is a diagonal n -by- n matrix in which the j 'th element is $1/k_{j,out}$ if $k_{j,out}$ is non-zero (the diagonal elements for which $k_{j,out} = 0$ simply do not matter).

Undirected networks are typically transformed to directed networks by replacing each undirected edge with two directed edges.

In practice, the computation of both Katz and PageRank centralities is performed numerically, using a power-iteration method that iterates the computation of the centrality values until those values converge. Typical values for $\frac{1}{\lambda}$ and β are 0.85 and $(1 - \frac{1}{\lambda})/n$, respectively – but as in the case of Katz centrality, it is theoretically possible that the PageRank centrality computation does not converge if λ is too low.

Food For Thought

Recall the concept of random walks on networks, that we introduced in Lesson-2. How can you interpret the PageRank centrality equation based on random walks?